

Blinded Visual Question-Answering

Evaluation of Qwen3-VL Plus on Hidden Gear Assemblies

Comprehensive methodology, performance, paired comparison, token, and cost report

Field	Value
Model	qwen3-vl-plus
Evaluation date	2026-08-11
Repository commit	360772e84d93f63010c806fc0faf21654bb8c4fc
Scope	28 source-hidden packages; hidden answer key kept local

Purpose: document methodology, exact-answer performance, question-type error rates, token use, estimated list-price cost, and per-item failures.

Prepared from retained provider responses, blind grading outputs, and independent reconciliation checks.

Abstract

This evaluation measured exact-answer performance of qwen3-vl-plus on 28 source-hidden packages from the Gear Assemblies QA repository. Each request contained one question and six rendered views in which configured exterior components were concealed. The answer key remained local and was not transmitted. Predictions were graded through a separate blind handoff. The model answered 8 of 28 items correctly, yielding 28.57% accuracy (95% Wilson CI, 15.3-47.1%) and a 71.43% error rate. Component-count accuracy was 75.00%, transmission-ratio accuracy was 12.50%, tooth-count accuracy was 0.00%, and the single gear-type item was correct. No hidden axis-angle item existed. The run used 116,838 tokens. Published list rates imply an estimated cost of USD 0.016984108, or about 1.70 cents.

28.57% Accuracy	71.43% Error rate	8 / 28 Exact answers	1.70 cents List-price estimate
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1. Introduction

Mechanical visual question answering requires recognition, unique-part counting, cross-view correspondence, geometric interpretation, and kinematic reasoning. The repository's hidden variants deliberately conceal configured exterior components while preserving the question and six-view presentation. This creates a visibility-stress condition for the same family of CAD assemblies.

The study used qwen3-vl-plus, Alibaba Cloud Model Studio's documented Qwen3 vision-language Plus alias [1]. The objectives were to quantify hidden-package exact-answer accuracy, characterize failures by question type, account for provider tokens and published-list-price cost, and compare the hidden run descriptively with the corresponding base-package predictions.

2. Methods

2.1 Dataset and hidden scope

The source was axioforgeai/gear-assemblies-qa at commit 360772e84d93f63010c806fc0faf21654bb8c4fc [4]. A shallow selector entered only the repository's source hidden directory and admitted directories matching ModelINN_NN_Hidden. It found exactly 28 packages. Every package contributed one prompt and six PNG views: isometric front-right-top, isometric front-left-top, isometric rear-right-top, front, top, and left. The evaluated set therefore contained 28 prompts and 168 images.

The selector rejected dot-prefixed, absolute, empty, traversal, and nested hidden paths. It copied only each package's prompt and referenced PNG files. The hidden ground-truth CSV, repository review files, base packages, and unrelated files were not copied into inference staging. The user explicitly authorized transmission of the 28 staged prompts and 168 staged images to Alibaba Cloud; the answer key was never transmitted.

2.2 Model request

Each hidden package was submitted independently to https://dashscope-us.aliyuncs.com/compatible-mode/v1, Alibaba Cloud's US OpenAI-compatible endpoint [3]. Requests used temperature 0, max_tokens 128, non-streaming output, and thinking disabled. The original question was followed by six image data URLs. To isolate the image variant, the system instruction matched the base evaluation exactly:

You are completing a blinded visual question-answering benchmark on a complete CAD gear assembly. Use only the question and six images in this request. Return exactly one JSON object with one key named answer. The value must be the shortest direct answer: a number only for numeric questions, or the requested category or term only. Do not include reasoning, units, markdown, or extra keys.

All 28 retained responses ended with finish reason stop. Every answer was nonempty, and no failed response entered the scored set. Retry metadata was not persisted, so the number of underlying attempts cannot be reconstructed from retained outputs.

2.3 Blind separation and independent audit

The inference-side subagent received only the hidden inference workspace and could inspect the staged prompts, six views per package, configuration, and retained outputs. It could not inspect ground truth, grading files, the base evaluation, the source repository, or dot-directories. Because the isolated agent lacked the trusted authorization transcript for an external data transfer, the transcript-bearing coordinator executed the fixed runner against the same staged workspace after explicit approval. The agent independently audited scope, model identity, response completion, usage arithmetic, and credential hygiene.

After all predictions were frozen, the coordinator created two grading files: one contained only package IDs and normalized predictions, and the other contained only package IDs and hidden answers. The grading subagent could read only that grading directory and could not inspect questions, images, raw responses, usage, the API key, source files, or inference outputs. It independently recomputed all 28 decisions. This was procedural blindness on a shared filesystem rather than an operating-system security sandbox.

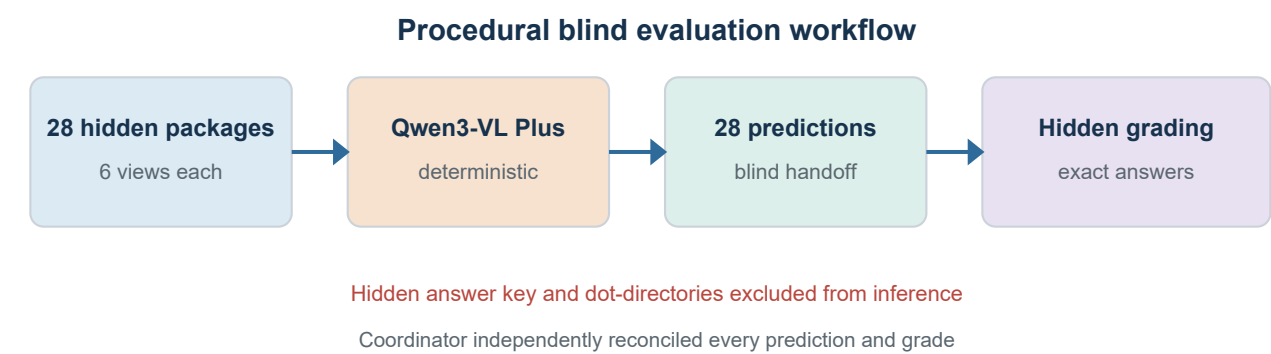


Figure 1. Hidden-package evaluation workflow. Staged prompts and images crossed the authorized API boundary; the hidden answer key remained in a separate local grading handoff.

2.4 Scoring and analysis

Numeric ground truth required one parsed numeric token equal to the expected value within absolute tolerance 1e-9. Discrete text was Unicode-normalized, stripped of surrounding whitespace and punctuation, lowercased, and matched exactly. Missing, ambiguous, unparsable, and failed predictions were defined as incorrect.

Accuracy was the fraction correct, and error rate was one minus accuracy. Wilson 95% confidence intervals describe binomial uncertainty. A post hoc taxonomy classified questions as transmission ratio, gear tooth count, component count, axis angle, or gear type. Mean absolute error and mean absolute percentage error were computed within numeric categories; percentage error excluded a zero-reference answer because division by zero is undefined. Paired comparisons used the 28 corresponding base questions. Two pairs had intentionally different hidden answer keys, so an additional same-target comparison excluded those pairs.

2.5 Token and cost accounting

Provider usage objects supplied token counts. Alibaba Cloud lists qwen3-vl-plus in the US or Global deployment at USD 0.143 per million input tokens and USD 1.434 per million output tokens for requests no larger than 32K input tokens [2]. The largest hidden request contained 4,183 prompt tokens, so every request stayed in that tier. The result is a published list-price estimate rather than a verified invoice; credits, discounts, taxes, account adjustments, and provider rounding may differ.

3. Results

3.1 Aggregate and category performance

The model returned 8 correct and 20 incorrect answers. Overall accuracy was 28.57% (95% Wilson CI, 15.3-47.1%), and error rate was 71.43%. Numeric accuracy was 25.93% (7/27); the single discrete gear-type item was correct.

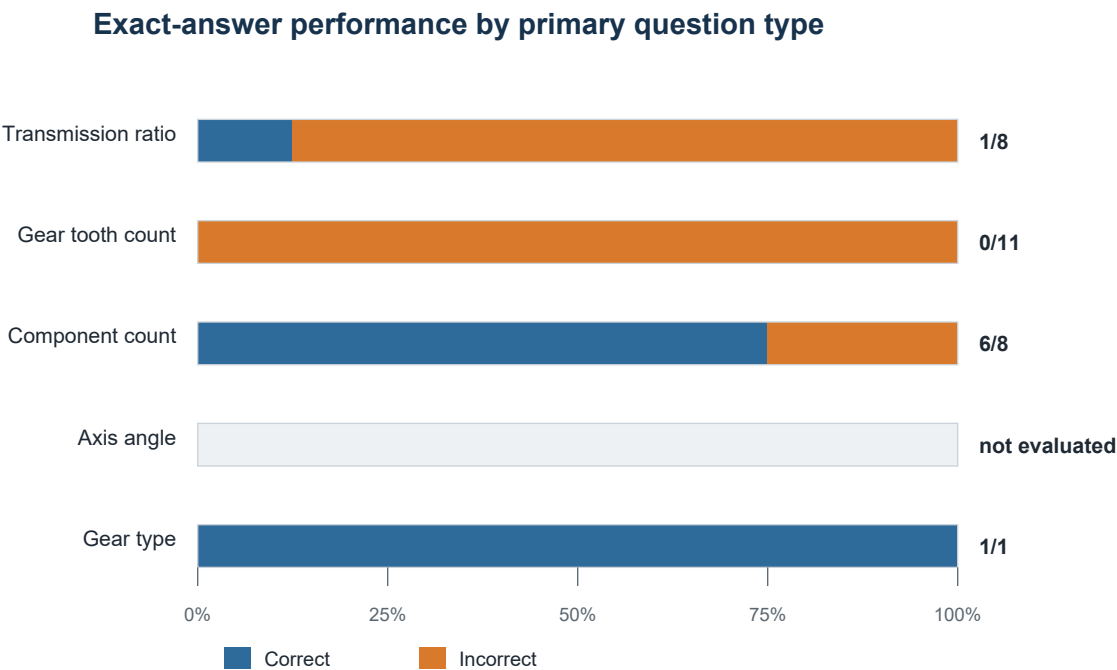


Figure 2. Exact-answer performance by primary question type. The axis-angle category was not represented among hidden packages.

Question type	n	Correct	Accuracy	Error	95% CI	MAE	MAPE
Transmission ratio	8	1	12.50%	87.50%	2.2-47.1%	5.475	40.65%
Gear tooth count	11	0	0.00%	100.00%	0.0-25.9%	16.364	38.75%
Component count	8	6	75.00%	25.00%	40.9-92.9%	0.375	17.86%
Axis angle	0	n/a	n/a	n/a	n/a	n/a	n/a
Gear type	1	1	100.00%	0.00%	20.7-100.0%	n/a	n/a

Table 1. Hidden-package accuracy, error rate, binomial uncertainty, and numeric error magnitude.

All 11 tooth-count questions were incorrect. Seven of eight transmission-ratio questions were incorrect. Component counting produced six correct answers in eight trials, and the single gear-type classification was correct. No hidden axis-angle question was available; its entries are therefore not applicable rather than zero performance. Confidence intervals remain wide because every category is small.

3.2 Subtype and numeric-error analysis

Question subtype	n	Correct	Accuracy	Error rate
Planetary ratio	2	0	0.00%	100.00%
Other train or gearbox ratio	6	1	16.67%	83.33%
Ring gear teeth	3	0	0.00%	100.00%
Sun gear teeth	3	0	0.00%	100.00%
Planet gear teeth	1	0	0.00%	100.00%
Smaller gear teeth	2	0	0.00%	100.00%
Larger gear teeth	2	0	0.00%	100.00%
Planet-gear quantity	3	3	100.00%	0.00%
Assembly inventory count	5	3	60.00%	40.00%
Gear-shape classification	1	1	100.00%	0.00%

Table 2. Descriptive performance by observed hidden-package question subtype.

All three planet-gear quantity questions were correct. Three of five broader assembly-inventory questions were correct. Every tooth-count subtype had a 100% error rate. One of six non-planetary gearbox-ratio items was correct, while both planetary-ratio items were wrong.

Of 27 numeric predictions, 15 were underestimates, 5 were overestimates, and 7 were exact. Among the 20 numeric errors alone, 75.00% were underestimates and 25.00% were overestimates.

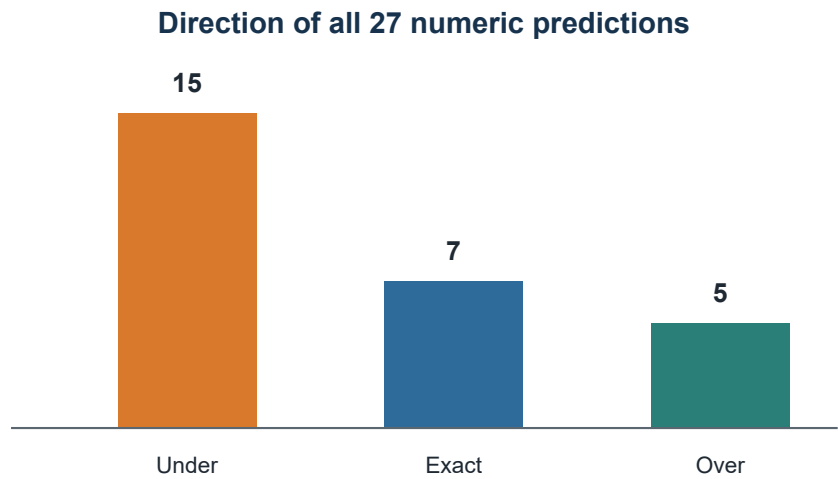


Figure 3. Direction of hidden-package numeric predictions relative to ground truth.

Transmission-ratio questions had MAE 5.475 and MAPE 40.65%. Tooth-count questions had MAE 16.364 teeth and MAPE 38.75%. Component counts had MAE 0.375; their MAPE of 17.86% excludes the zero-reference item.

3.3 Token use and estimated cost

Measure	Observed value
Successful requests	28
Prompt tokens	116,624
Image tokens	113,736
Text prompt tokens	2,888
Cached tokens	0
Completion tokens	214
Total tokens	116,838
Prompt tokens per request	4,160 min; 4165.14 mean; 4,183 max
Completion tokens per request	7 min; 7.64 mean; 9 max
Input list-price estimate	USD 0.016677232
Output list-price estimate	USD 0.000306876
Total list-price estimate	USD 0.016984108 (about 1.70 cents)
Estimated cost per question	USD 0.000606575
Estimated cost per correct answer	USD 0.002123014

Table 3. Provider-reported hidden-run token usage and published list-price estimate.

The 28 retained completion timestamps spanned 81.7 seconds. This interval reflects sequential response collection and excludes staging, approval, auditing, grading, and reporting.

3.4 Paired comparison with base views

Across all 28 matched question identifiers, the corresponding base run answered 4 correctly (14.29%), whereas the hidden run answered 8 correctly (28.57%). Twenty pairs were wrong in both runs, four were correct in both, four were correct only in the hidden run, and none were correct only in the base run. This descriptive difference is not evidence that hiding components improves performance.

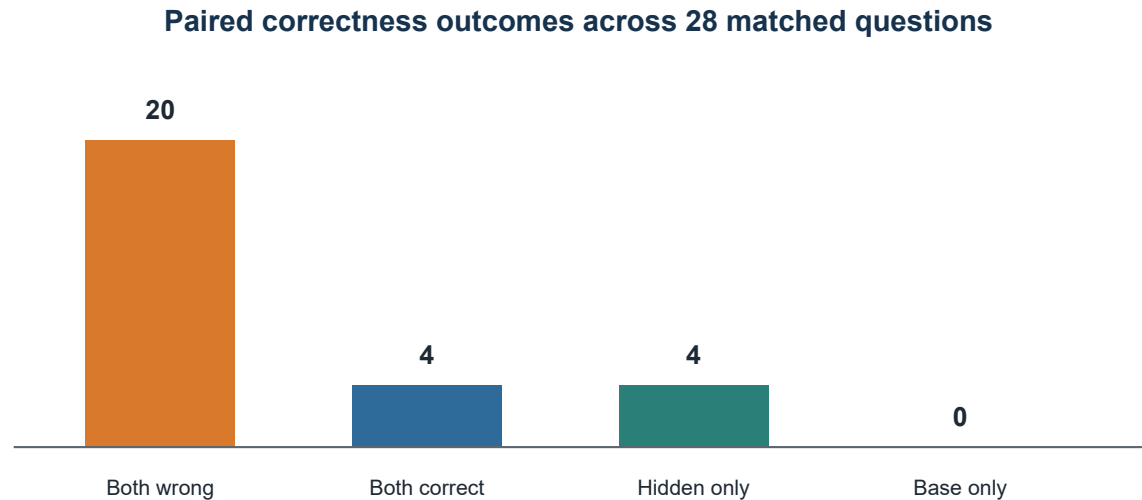


Figure 4. Correctness transitions for the 28 hidden/base question pairs.

Question type	Paired n	Base correct	Hidden correct	Difference
Transmission ratio	8	0/8	1/8	+12.50 pp
Gear tooth count	11	0/11	0/11	+0.00 pp
Component count	8	4/8	6/8	+25.00 pp
Gear type	1	0/1	1/1	+100.00 pp

Table 4. Descriptive accuracy differences across matched hidden and base question identifiers.

The source answer key intentionally changes for Model02_02_Hidden, Model02_03_Hidden. Restricting the comparison to the 26 pairs with identical targets yielded 3/26 correct for the base run and 6/26 for the hidden run. Three same-target pairs improved and none regressed; an exact two-sided McNemar test for three discordant pairs gives $p = 0.25$. The sample therefore does not establish a reliable visibility effect.

3.5 Correct-answer control cases

Package	Type	Question	Correct	Predicted
Model02_02_Hidden	Component count	How many support brackets are there in this assembly?	0	0
Model02_03_Hidden	Component count	How many distinct parts are there in this assembly?	4	4
Model03_04_Hidden	Component count	How many planet gears are there in this planetary gear train?	3	3
Model04_05_Hidden	Component count	How many planet gears are there in this planetary gear train?	3	3
Model07_02_Hidden	Component count	How many shafts are there in this gear box?	3	3
Model07_03_Hidden	Transmission ratio	What is the transmission ratio of this gear train?	12	12
Model08_04_Hidden	Gear type	What is the type of the gear shape? Spur, Helical, Bevel or Double Helical or others.	Helical	Helical
Model09_04_Hidden	Component count	How many planet gears are there?	4	4

Table 5. Eight correct hidden-package items retained as controls for extraction and grading.

4. Discussion

The hidden run remained weak on exact mechanical inference. Every tooth-count item failed, and seven of eight transmission-ratio answers were wrong. Those 18 errors accounted for 90% of all failures. The images and direct-answer prompt did not provide a reliable path from visible geometry to exact enumeration or kinematic ratios.

Component counting was stronger. All planet-gear quantity items and three of five broader inventory items were correct. The single gear-shape classification also changed from wrong in the base run to correct in the hidden run. Concealment may sometimes remove clutter or emphasize remaining geometry, but a single deterministic request per variant cannot identify that mechanism.

The hidden run outscored the matched base subset without any observed correctness regression. This result is counterintuitive if concealment is assumed to reduce evidence. However, two hidden targets differ from their base answers, the sample contains only three same-target discordant improvements, and service-level generation can remain nondeterministic even at temperature 0. Repeated paired trials are required before attributing the difference to visibility.

5. Limitations and recommended follow-up

This benchmark covers one model alias, one prompt, one run per variant, one commit, and 28 hidden packages. No hidden axis-angle or worm-ratio item existed. The taxonomy was post hoc, exact-match grading awarded no partial credit, reasoning was suppressed, and retry counts were not recorded. Errors therefore cannot be decomposed into perception, cross-view matching, enumeration, kinematic setup, arithmetic, or answer selection.

A stronger study would pin a dated model snapshot, repeat every base/hidden pair, pre-register the paired analysis, separate pairs with changed targets, and preserve attempt metadata. Targeted crops or an explicit tool-assisted counting and ratio-calculation condition could test whether failures arise from visual resolution or unsupported mental arithmetic.

6. Conclusion

Under the authorized hidden-view protocol, qwen3-vl-plus achieved 28.57% exact-answer accuracy and a 71.43% error rate. Component and gear-type questions were substantially stronger than tooth counts and transmission ratios. The hidden run descriptively exceeded the paired base subset, but the small discordant count, changed targets, and absence of repeated trials prevent a causal conclusion.

Data availability and audit trail

The local hidden-evaluation directory retains staged prompts and views, 28 raw provider responses, normalized predictions, run state, inference audit, blind grading handoffs, grading outputs, report sources, and cost inputs. Independent inference and grading audits matched all IDs and arithmetic. Retained output scans found no credential-shaped values. The answer key remained local throughout inference.

Security and operational note

The transfer began only after explicit user authorization identified the Alibaba Cloud destination and the 28-prompt, 168-image payload. The runner parsed credential candidates line by line, retained only the successful candidate ordinal, sanitized errors, and preserved TLS verification. No credential string appeared in retained outputs.

References

- [1] Alibaba Cloud. Qwen-VL compatible with OpenAI. Accessed Aug. 11, 2026. <https://www.alibabacloud.com/help/en/model-studio/qwen-vl-compatible-with-openai>
- [2] Alibaba Cloud. Model Studio model pricing. Accessed Aug. 11, 2026. <https://www.alibabacloud.com/help/en/model-studio/model-pricing>
- [3] Alibaba Cloud. API endpoint. Accessed Aug. 11, 2026. <https://www.alibabacloud.com/help/en/model-studio/base-url>
- [4] axioforgeai. gear-assemblies-qa. GitHub commit 360772e84d93f63010c806fc0faf21654bb8c4fc. <https://github.com/axioforgeai/gear-assemblies-qa/tree/360772e84d93f63010c806fc0faf21654bb8c4fc>